Pham Tran Thanh Tai

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Research Interests

Trustworthy and explainable machine learning for data-driven decision making: explainable AI (SHAP/LIME-style local explanations), statistical machine learning, and predictive modeling on real-world industrial and business data. Keen to join the TrustAIM Lab (Trustworthy AI & Intelligent Manufacturing) at NYCU to learn uncertainty-aware learning, XAI, and manufacturing data science, and to contribute through hands-on data engineering and modeling.

Education

University of Economics and Law – VNU-HCM (UEL) Ho Chi Minh City, Vietnam
B.S. in Digital Business and Artificial Intelligence; GPA: 8.36/10 (Very Good, cumulative) Expected: May 2027

- Relevant coursework: Machine Learning, Time-Series Analysis & Forecasting, Advanced Data Analytics, Database Systems, Data Warehousing & Integration, AI Solutions for Business & Management

Experience

TLC – Coffee Market Quantitative Research Platform July 2026 - September 2026
Data / Quantitative Analyst Ho Chi Minh City, Vietnam (Hybrid)

- Built Python data pipelines ingesting heterogeneous broker sources (PDF, HTML, spreadsheets), cleaning and normalizing them into analysis-ready datasets in PostgreSQL for time-series and market analysis.
- Added an LLM-based layer (LangChain/LangGraph) for topic classification and sentiment scoring of market news, turning unstructured text into structured signals; orchestrated daily refreshes with Apache Airflow.

THD Cyber Security JSC – R&D Dept. August 2025 - October 2025
Business Intelligence / Business Analyst Intern Ho Chi Minh City, Vietnam

- Gathered requirements and defined KPIs for Sales & HR dashboards in a Data Warehouse project, producing BRD documentation, source-to-target data mapping, and UAT test cases.
- Modeled and transformed source data into a star-schema warehouse and built the ETL/data-prep layer feeding the reporting dashboards.
- Delivered interactive dashboards that streamlined reporting and enabled data-driven decisions for Sales and HR stakeholders.

Projects

Golden Wheat JSC – Digitalization of Data Entry & Reporting (Outsourcing) Solution Architecture phase

- Designing the solution architecture to move the client's reporting data from manual entry to automated data entry, with a central database and Metabase dashboards for reporting.
- Scope so far: solution design only; implementation has not started.

Machine Learning-based Prediction of Late Delivery Risk (with SHAP explainability) Research Lead

- Led a supervised-learning study on supply-chain risk: engineered features, trained and compared classifiers, evaluated with cross-validation and standard classification metrics.
- Applied SHAP to explain which factors drive predicted risk, making black-box model output interpretable for decision makers; authored the report and defended it before a faculty review board (Python: scikit-learn, pandas).

Traceability & Management System for Ca Mau Crab Supply Chain Graded A+, Team Lead

- Designed a relational database with ERD and 3NF normalization; wrote complex SQL using CTEs and window functions for analytical reporting queries.

MIS Final Project: Big Data Solutions Analysis for Viettel Graded A, Team Lead

- Researched Big Data infrastructure and analytics platforms for customer-service optimization and fraud detection.

Skills

Machine Learning & XAI:	Python, scikit-learn, pandas, NumPy, feature engineering, cross-validation, SHAP
Statistics & Forecasting:	Time-series analysis & forecasting, clustering, statistical modeling
Data & Pipelines:	SQL (PostgreSQL, SQL Server), ETL, data modeling, Apache Airflow, data warehousing
LLM & Visualization:	LangChain/LangGraph, LLM APIs, Power BI, DAX, Power Query (M)
Certifications:	Business Analysis & Process Management, IBM Data Analyst (Coursera)
Languages:	English – TOEIC L&R 835, S&W 260 (~CEFR B2); Vietnamese – Native